5.3 Minimum Requirements for TCPs

5.3.1 Physical and Mental Requirements

TCPs should be physically and mentally prepared to do the required work, and should exhibit these characteristics:

- good vision
- good hearing
- alertness and mature judgement
- intelligence and common sense
- pleasant, cooperative disposition
- sense of responsibility for the safety of workers and the public

5.3.2 Training and Certification

TCPs shall receive approved training, pass an examination, and be certified before they are assigned to work within a work zone.

TCPs shall:

- have valid proof of training or certification issued by a recognized training agency as determined by WorkSafeBC
- carry their certification at all times while on the job
- present their certification to the appropriate authorities on demand

TCPs should have these proficiencies:

- knowledge of vehicle operations, such as stopping distances, turning radii, etc.
- knowledge of basic regulatory requirements governing drivers
- understanding of basic traffic signs and the rules of the road
- understanding of driver expectations

A Class 5 driver's licence may help to provide information about these proficiencies.

5.4 Work Zone Apparel and Equipment

5.4.1 TCP Apparel and Equipment

Personal protective clothing and equipment for TCPs shall comply with Parts 8 and 18 of WorkSafeBC's Occupational Health and Safety Regulations and other standards as identified below.

1. Basic Requirements

TCPs shall have the following required material with them on the job at all times:

- **STOP or SLOW C-027 Paddle:** An extension pole that is 1.3 to 2.1 metres long is optional.
- **Traffic Control Person Ahead C-001-1 Sign:** The sign shall be removed or covered when TCPs are not actively controlling traffic.
- **Safety Headgear:** TCP hard hats shall comply with one of these standards:
  - Current CSA Standard CAN/CSA-Z94.1 Industrial Protective Headwear
  - Current ANSI Standard Z89.1, American National Standard for Personnel Protection – Protective Headwear for Industrial Workers Requirements
  - Current Japanese Industrial Standard JIS T 8131, Industrial Safety Helmets for Class AB or ABE Headgear

  Hard hats shall be of a **high-visibility colour** with a band of retroreflective tape across the top from front to back and on the sides.
  - WorkSafeBC permits fluorescent yellow-green, fluorescent orange-red, and fluorescent red colours.
  - The hard hat shall have retroreflective material across the top from front to back and on the sides to make it clearly visible to drivers approaching from any angle.
- **Safety Footwear:** TCP footwear shall be CSA-compliant Grade 1 safety footwear (green triangular CSA patch on the outside, green rectangular label on the inside).
- **Safety Apparel:** TCPs shall wear Class 3 garments that comply with both the current CSA Z96 standard and Section 18 of WorkSafeBC's Occupational Health and Safety Regulation.

See also *Section 5.4.2: Apparel Retroreflectivity for TCPs*.



Figure 5.1: C-027 Traffic Control Paddle STOP or SLOW – Double Sided

2. Day-Time and Night-Time Apparel

Alternative to coveralls shown, TCPs may wear a combination of a torso vest (or jacket) and bands encircling both arms and both legs.

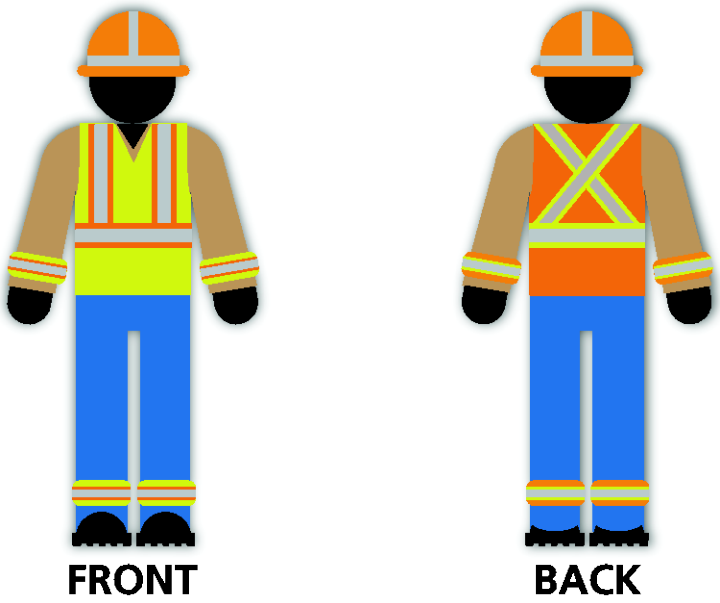


Figure 5.2 A: TCPs with Fluorescent Yellow-Green and Fluorescent Orange-Red Vests

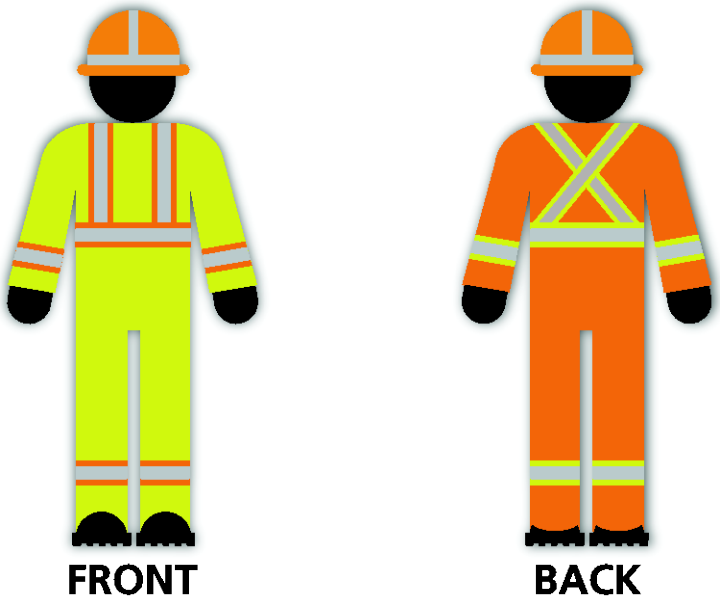


Figure 5.2 B: TCPs with Fluorescent Yellow-Green and Fluorescent Orange-Red Coveralls

3. Additional Requirements for Night Operations

- flashlight with red signalling wand
- spare batteries
- two-way radios

4. Night Lighting

TCP stations shall be illuminated at night. If street lighting is available, TCPs should stand below the light to maximize front-of-body illumination. If temporary overhead lighting is being used, it shall not subject approaching drivers to excessive glare.

5. Optional Equipment

- CSA-approved safety sunglasses or eye protection where required
- rain gear meeting Class 3 retroreflectivity requirements

6. Lettering and ID Patches

Lettering and ID patches that are not retroreflective may be placed on the garment, provided that they do not cover an area greater than 105 cm^2 and do not cover any part of the mandatory retroreflective stripes/bands.

Retroreflective lettering or ID patches that meet the requirements of the current CSA Z96 standard shall not cover an area greater than 500 cm^2 and may be placed anywhere on the garment as long as the positioning does not obscure the recognizable pattern of the stripes/bands.

Note: For work on Provincial roadways, the Ministry has adopted a high-visibility standard for worker apparel that exceeds WorkSafeBC's retroreflectivity standard.

This standard provides both adequate retroreflectivity and contrasting colour, and applies to all workers on all Ministry projects.

5.4.2 Apparel Retroreflectivity for TCPs

TCPs shall wear Class 3 safety garments that comply with the current CSA Z96 standard and the WorkSafeBC requirement. At minimum, Class 3 high-visibility material shall fully cover the upper torso (front, back, sides, and over shoulders) and shall include bands encircling both arms and both legs.

Acceptable colours for background material on these high-visibility safety garments are **fluorescent yellow-green** and **fluorescent orange-red** (the orange-red is often labelled fluorescent orange).

These garments require a **contrasting-colour fluorescent** stripe that is at least 100 mm (4") wide. Acceptable colours for the contrasting stripe are also fluorescent yellow-green and fluorescent orange-red. The retroreflective bands used on these garments shall be at least 50 mm (2") wide and in a colour that contrasts with the background colour.

Horizontal wrist and ankle stripes/bands shall be placed on the sleeves and pants, encircling both arms and legs. They shall be 100 mm (4") wide and include a 50 mm (2") retroreflective band with two 25 mm (1") contrasting colour fluorescent stripes on each side of the retroreflective band.

The stripes/bands shall be laid out in this pattern:

- symmetric **X** on back of garment extending from shoulders to waist
- two vertical stripes/bands on front extending over shoulders and down to waist
- horizontal leg and arm stripes/bands encircling both arms and both legs.
- waist-level, horizontal stripe/band extending entirely around the circumference of the torso from the back to the bottom of the vertical stripe/bands on the front, where they end at the front fastening mechanism (snap, zipper, etc.)
- gaps in retroreflective materials for front fastening cannot exceed 50 mm (2")

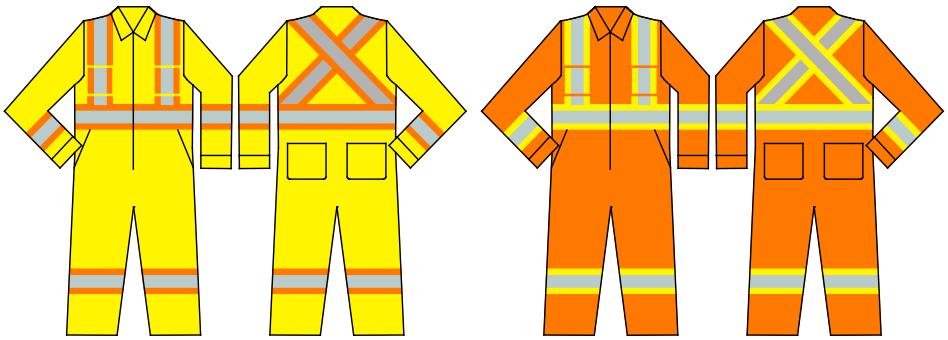


Figure 5.3: Coveralls with Contrasting Retroreflective Bands

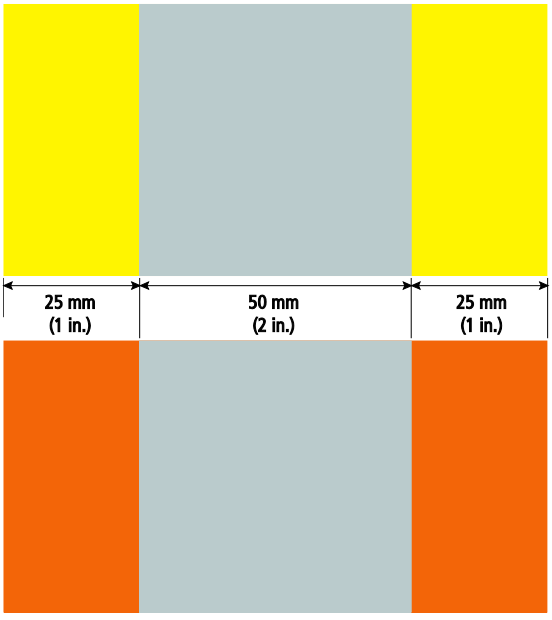


Figure 5.4: Retroreflective Striping Cross Sections and Dimensions

5.4.3 Apparel for Other Onsite Workers

Work zone workers who are not TCPs shall wear Class 2 safety garments that comply with both the current CSA Z96 standard and the WorkSafeBC requirement.

At minimum, Class 2 high-visibility material shall fully cover the upper torso (front, back, sides, and over the shoulders).

For work on Ministry right-of-way, onsite workers shall wear safety garments that comply with the standards outlined in this Manual, the current Z96 standard, and current WorkSafeBC Part 8 requirements.

The following apparel components shall also comply with the retroreflectivity requirements for TCPs (see *Section 5.4.2: Apparel Retroreflectivity for TCPs):*

- fluorescent background material
- fluorescent 100 mm (4”) contrasting stripe
- 50 mm (2”) retroeflective bands of tape



Figure 5.5: Worker Vests with Contrasting Retroreflective Bands (Day-Time Apparel for TCPs)

Note: For work on Provincial roadways, the Ministry adopted standard outlined above for work apparel exceeds WorkSafeBC’s standard.

This standard provides both adequate retroreflectivity and contrasting colour, and applies to all workers on all Ministry projects.

5.4.4 Apparel Labelling for All Onsite Workers

Garment labels should include these details to comply with the current CAN/CSA Z96 standard:

1. Manufacturer or authorized representative name, trademark, or other form of identification.
2. Designation of the product type (i.e., Coverall), commercial name, or code.
3. Size designation.
4. CSA Z96.
5. Apparel Class and Level of Performance for the retroreflective material.
6. Indication that background material is fluorescent.
7. Indication of Flame Resistant (FR) Performance if applicable.

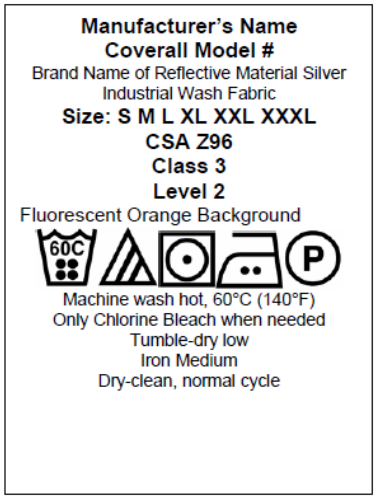


Figure 5.6: Manufacturer's Label

5.5 TCP Communications

5.5.1 Fundamental Principles

TCPs work together to regulate traffic through the work zone. This means that they need to communicate effectively with each other.

When the two TCPs are **within sight of each other:**

- They should use pre-arranged visual signals to communicate.
- One TCP should wait until signals are acknowledged by the other TCP before changing traffic flow.

When the two TCPs are **not inter-visible,** such as on curves or hills, they should either use two-way radios or take the following steps:

1. Station a third TCP between them so that signals can be relayed visually. This third person should stand outside the travel lanes at a location visible to the two other TCPs. This will be practicable only within short work zones.

   For illustrations of these positioning requirements, see *Figure 5.7: Positioning Requirements When Two TCPs Are Not Inter-Visible*.
2. Equip the intermediate TCP with a Stop/Slow paddle for relaying signals from the TCP at one end to the TCP at the other end.
3. Ensure that all three TCPs understand and acknowledge the pre-arranged signals.

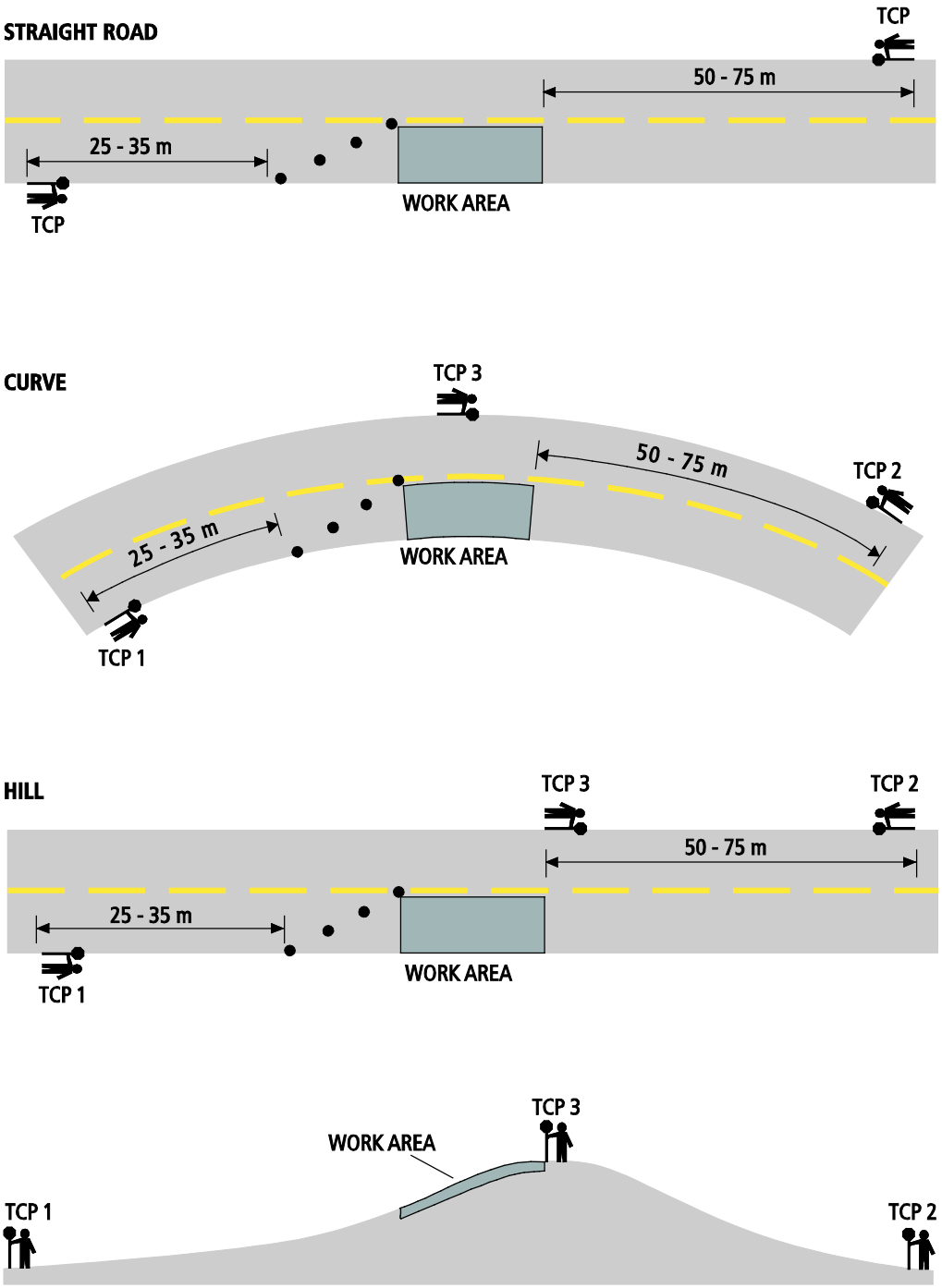


Figure 5.7: Positioning Requirements When Two TCPs Are Not Inter-Visible

5.5.2 TCP Radios

1. Radio-Based Tasks

Using radios allows TCPs to carry out several important tasks:

- communicate with others in areas where they cannot be seen
- pass along information about traffic in the queue or passing through the site
- advise of movements or encroachment of construction traffic on travelling lanes
- smoothly coordinate the movement of public and construction traffic
- advise of incidents or issues that may occur on the site
- warn of approaching emergency vehicles
- coordinate safe passage of construction traffic past the vehicle queue

2. Rules for Radio Use

When using radios, TCPs should follow these rules:

- use as specified in the manufacturer's instructions and conditions of use
- ensure that radios work properly across the work zone or work activity area before beginning to use them for traffic control
- carry spare batteries and have chargers readily available
- use only one ear for a headset or receiver, keeping the other free for hearing other noises in the area
- ensure that both hands are free for use
- pre-arrange voice signals for every situation, and do not change them
- speak clearly
- ask for unclear messages to be repeated
- avoid unnecessary talk
- avoid inappropriate comments
- be aware of signal delay and allow time for the transmission to get through

3. Radio Frequencies

It is important to use radio frequencies that allow for communication not only between TCPs but also with the Site Foreman, First Aid Attendant, and equipment operators who may be encroaching on or entering into the travel lanes.

It is best to use one common frequency for all onsite personnel. If this is not possible, the Traffic Control Supervisor should carry an additional radio that uses the work site frequency in order to communicate with the work site and help to coordinate movements and pass information to TCPs on their radio frequency.

5.6 TCP Positioning and Signals

5.6.1 Hazard and Risk Assessment

When deciding on a position for the TCP and the traffic queue, it is important to identify and assess the potential risks associated with all site hazards.

If TCPs and traffic queues will be positioned such that the TCPs are at high risk from a hazard, appropriate steps should be taken to eliminate or minimize the risk. It may be necessary to remove the hazard or reposition the TCP.

Hazards that create risk for TCPs include, but are not limited to:

- rock fall areas or avalanche zones (seasonal)
- blind corners and hill crests
- tunnel entrances and exits
- lengthy or steep grades
- danger trees
- wildlife
- dark or remote areas
- heavy traffic congestion
- large commercial or business accesses

Note: TCPs should be able to focus their attention on traffic and not be distracted by having to watch out for other hazards.

The distance between the Traffic Control Person Ahead C-001-1 sign and the TCP should not exceed 150 metres unless local site conditions (curves, hills, etc.) govern. If there is not an ideal location within this distance because of road features or conditions, an additional sign should be used in advance of the C-001-1, such as a Flagger Ahead C-001-2 sign or a Prepare to Stop C-029 sign.

5.6.2 Positioning Rules for TCPs

1. Stand either on the shoulder adjacent to the traffic being controlled or in a lane that has been closed to traffic, on the same side of the roadway where you are controlling traffic. Be aware that the closed lane is not the opposing lane, even when controlled by another TCP or device.
    - Always plan an escape route from every position you assume—i.e., an uninhibited path for avoiding errant vehicles (see also *Section 5.7.1: Ability to Make Evasive Manoeuvres*).
    - After more than one vehicle has been stopped—and only if necessary—you may move into the lane under your control to assess queue length or to achieve a better view of approaching vehicles.
    - Avoid entering a lane being used by opposing traffic.
    - Return to your starting position before you release the stopped traffic queue.

2. Unless otherwise specified, stand 25 to 35 metres from the TCP taper and 50 to 75 metres from the downstream taper to avoid out-of-control vehicles and to provide manoeuvring room for responding to vehicles that make unanticipated lane changes.
3. Face the centre of the road, with your back to the road shoulder, scanning traffic approaching from both directions. Remain aware of what is happening in the stopped lane.
4. For intersection traffic control, it may be necessary to stand in the middle of the intersection, in which case it may not be possible to comply with the three rules above.
5. Stand where you can see equipment on the site and where you can see—and be seen by—approaching drivers. To the extent practicable, stand where the background will make you as conspicuous as possible.
6. To be visible to drivers, stand away from the other workers, and never stand in a group of people while stopping traffic.
7. Never use your body as a barrier for blocking errant vehicles.

Regardless of the rules listed above, TCP safety is paramount. Therefore, always stand where you can see and be seen by approaching drivers, in a position that is suitable for safely stopping traffic and/or directing traffic through the work activity area, and where there is an escape route.

5.6.3 Positioning Rules for TCPs in Intersections

1. TCP direction in intersections cannot conflict with the direction provided by any existing intersection control. Traffic signals shall be shut off or changed to flash mode. Stop signs shall be covered.
2. Traffic Control Persons must be visible to approaching traffic and not obscured by advance warning or other signage.

5.6.4 Temporary Stop Bars

Temporary stop bars (stop lines) may be used by TCPs to help define a specific stopping location in advance of the TCP position.

Temporary stop bars not only provide road users with defined stopping locations but also help TCPs to maintain a safe separation from stopped vehicles.

A temporary stop bar must be white, and at least 25 cm (10") wide. It should extend across the full width of the lane for which it is intended, perpendicular to the direction of vehicular travel. It is made of low-profile plastic or another temporary material that is heavy enough not to be displaced when vehicles stop on it or are driven over it at anticipated speeds. The profile must be low enough that it does not impede traffic flow when being crossed by traffic.

One tubular marker should be used on the shoulder side and one on the median side of a temporary stop bar, with each tubular marker displaying the appropriate Stop Line R-025-R or R-025-L sign.



Figure 5.8: Temporary Stop Bar

A temporary stop bar should be placed at least one-half of Distance A from the TCP (Distance A values are those shown for Construction Sign Spacing in *Table B – Device Spacing Lengths*, see *Section 6.6 or Appendix F*), with sufficient sight distance provided for approaching drivers.

A temporary stop bar should not be placed:

- on sharp horizontal or vertical curves
- across bicycle lanes
- through pedestrian crossings
- within intersections that have permanent stop bars

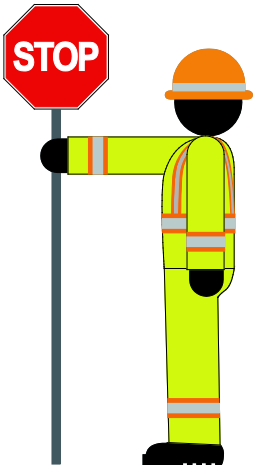
5.6.5 TCP Signals

TCP signals shall comply with the specifications described and illustrated in Part 18 of WorkSafeBC's Occupational Health and Safety Regulation:

http://www2.worksafebc.com/publications/ohsregulation/part18.asp

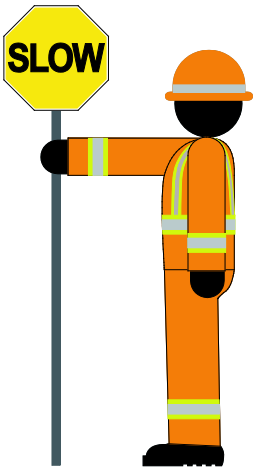
1. Signal for Stopping Traffic

1. Position yourself in a safe position. See *5.6.2 Positioning Rules for TCPs*.
2. Stand on the roadway shoulder, where you can see, and be seen, with toes pointing towards the centre of the road, and hold the paddle out to stop the first vehicle. Always display the paddle in a static manner, and hold the paddle so that it is visible to traffic.
3. Once you have stopped the first vehicle, adjust your position so that you are standing in a position where you can see, and be seen by, approaching drivers from a sufficient distance to stop safely (at least 150 metres).



2. Signal for Slowing Traffic

1. Extend the traffic control paddle towards the lane of oncoming traffic.
2. Wave the traffic forward with your other hand to avoid bringing traffic to a full stop.



3. Stopping Sight Distances

When slowing and stopping traffic, it is critical to remember the required stopping distances for vehicles travelling at various speeds. The faster a vehicle is moving, the more distance it requires to stop. The size and weight of a vehicle also affect its stopping distance.

Table 5.1: Stopping Sight Distances

Vehicle Speed (km/h)	Stopping Sight Distance (m)	Notes
50	60	• These are stopping sight distances (SSD) for passenger vehicles in wet conditions on a level roadway. • More stopping distance is required for larger, heavier vehicles. • More stopping distance is required on a downgrade. • The table values are from the Transportation Association of Canada (TAC) Geometric Design Guide for Canadian Roads (2017), rounded to the nearest 10.
60	80	
70	110	

5.7 TCP Safety

5.7.1 Ability to Make Evasive Manoeuvres

1. TCPs should be positioned so that they can make evasive manoeuvres to avoid being struck by a vehicle.
2. If an errant vehicle enters the work site, TCPs are responsible for using their escape routes.
3. If it appears that the vehicle is not stopping, the TCP should notify personnel working on the site (via radio or audible device), and observe and document as many details as possible for subsequent follow-up.
4. Although many TCPs use their vehicles as refuge, they should not position themselves in such a way that the presence of the vehicle or other equipment reduces their options for making evasive manoeuvres.

   The same precautionary principle applies to working near equipment, barriers, or opposing traffic.

5.7.2 Management of Approaching Vehicle Speeds

1. Speed Management Delineation

Centreline or edge line delineation can be used to reduce vehicle speed in advance of the TCP position.

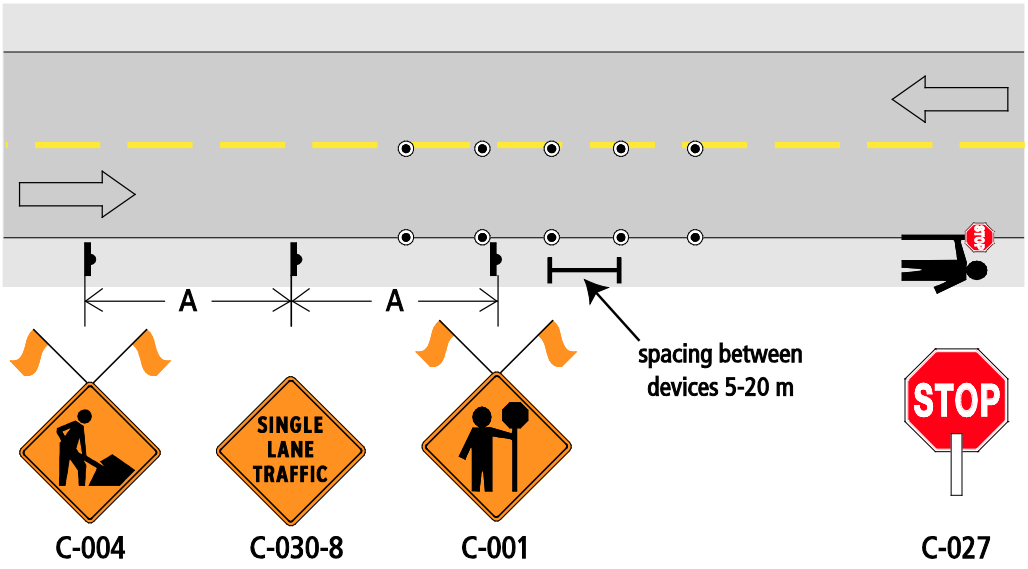


Figure 5.9: Centreline and Edge Line Delineation

2. Speed Management at Night

When TCPs are working during hours of darkness and are having difficulty getting traffic to stop, various traffic control options may assist with slowing and stopping traffic.

Examples include, but are not limited to:

- increasing number of advance warning signs and devices
- increasing sign sizes
- using dynamic messaging signs (DMS)
- using speed reader boards
- using an oversized STOP R-001 sign (75 cm x 75 cm) on a barricade across the closed lane where the TCP is positioned

5.7.3 Prohibitions for TCPs While Actively Controlling Traffic

1. Never stand near a vehicle or sit in a vehicle when actively controlling traffic.
2. Never argue with a driver.
3. Never stand in an open travelled portion of the roadway while traffic is moving.
4. Never accept an assignment to carry out other onsite work, and never attempt to carry out any other onsite work.
5. Never allow the TCP sign to be displayed when a TCP is not directing traffic.
6. Never give direction that contradicts a traffic signal.
7. Never converse with any person about anything that is not work-related, and ensure that all work-related conversation is both necessary and brief.
8. Never sit when actively controlling traffic.
9. Never lean on a post or other object.
10. Never use a mobile device, tape, disk, MP3 player, TV, non-work radio, or any other device that impairs sight, hearing, or attention. Use cell phones only to communicate about onsite emergencies.
11. Never stand near equipment.
12. Never turn your back on approaching traffic.
13. Never wear clothing or items that can obscure or reduce peripheral vision, such as hoodies, certain kinds of sunglasses, etc.
14. Never become impatient or enraged.
15. Never attempt to slow traffic by displaying the STOP sign rather than the SLOW sign.
16. Never leave the control position without being replaced. Meal, coffee, toilet, and rest breaks should be pre-arranged before work starts.
17. Never regulate traffic if your judgment is impaired in any way.
18. Never regulate traffic if you have suffered a reduction in performance that could increase anyone's exposure to risk.

5.8 Emergency Procedures

5.8.1 Passage of Emergency Vehicles and Personnel

TCPs should review the Traffic Management Plan, which may specify how emergency vehicles and personnel are to be accommodated or taken through the work zone, and should discuss the process to be used in these situations with the Traffic Control Supervisor and Site Supervisor.

TCPs need to be aware of any instructions that should be communicated to the drivers of emergency vehicles, including:

- the path to drive
- where hazards may exist
- any communications required along the way (e.g., the lead vehicle may be given a radio with the site frequency to be returned to the TCP at the other end)
- a site map if the work zone is long, such as a repair work zone established to deal with a significant flood event

In smaller rural communities, many emergency service providers are volunteers, which means that the emergency facility is not staffed on a regular basis. In these situations, volunteers may be driving personal vehicles to their “hall” to pick up emergency vehicles.

There should be a process for allowing these volunteers to get to their hall quickly, keeping in mind that they may be passing through the work zone again shortly thereafter with an emergency response vehicle.

5.8.2 Traffic Control at Emergency Scenes

Members of emergency services and recovery groups that may respond to a motor vehicle incident may include:

- police, fire, and ambulance responders
- highway rescue and search and rescue responders
- towing companies
- Road Authority officials
- maintenance contractors
- other emergency groups

Members of these emergency responder groups often have to control traffic around the site of an emergency or crash. Before implementing traffic control, responders and workers should ensure their own safety and the safety of others.

Having assured the safety of themselves and other onsite personnel, emergency responders are expected to maintain traffic operations through the area impacted by the emergency by employing basic traffic control principles, and to be trained in:

- basic traffic control techniques
- traffic control equipment setup, operation, and take-down
- the traffic management principles outlined in this Manual
- the use of a buffer vehicle to protect the workplace
- the use of appropriate personal protective clothing and safety equipment
- other appropriate safe work procedures

If the traffic control situation will persist for more than two hours, the emergency responders directing traffic are expected to be trained in a manner acceptable to WorkSafeBC for high-risk traffic control or to be replaced by personnel who have this training.

PART B – TRAFFIC CONTROL

Section 6: Traffic Control Layouts – General Instructions

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PART B – TRAFFIC CONTROL

Section 6: Traffic Control Layouts – General Instructions

This section outlines the appropriate development and use of the traffic control layouts described in Sections 7 to 19. The user of this manual should have a thorough understanding of the information in this section before applying the layouts.

Important Notes for Layouts in Sections 7 to 19

1. The traffic control layouts are considered the minimum standard. The associated text description highlights the key standards as well as guidance and options that can be considered by the user.

2. Although this Manual often identifies traffic control requirements by using the word "shall," there may be circumstances where strict compliance with the requirements is not reasonable and it will be necessary to deviate from the requirements. In these cases, written justification for the modification must be recorded and depending on the complexity of the modification, written permission may need to be granted by the Road Authority.

3. A variety of conditions may be encountered, and no single standard sequences of signs or other traffic control devices can be set up as an inflexible arrangement for all conditions and locations. Furthermore, it may be impossible to comply with the minimum standards in emergency situations.

4. If the layouts in the Manual are not working in a particular situation, alternatives should be considered and implemented.

5. If it is necessary to deviate from the requirements in the Manual, the Prime Contractor should contact the Road Authority.

TCP Acronym

TCP is generally used for Traffic Control Persons throughout Sections 6 to 19.

Side-by-Side Formatting

The layouts in Sections 7 to 19 are shown with the **description on the left** and the corresponding **diagram on the right** so that readers can see them side-by-side

6.1 Introduction to Work Zone Components

Temporary traffic control measures are required within work zones to safely guide and protect road users and workers.

For the purposes of this Manual, a **work zone** is:

> A roadway area in which road user conditions have been temporarily changed for the purpose of construction, maintenance, utility work, or for emergency incident management.

A work zone is typically marked by temporary traffic control devices such as signs, channelizing devices, barriers, pavement markings, and/or work vehicles. It usually extends from the first to the last of the traffic control devices, and is generally the area between the first advance warning sign and a point beyond the work operations where traffic is no longer affected by temporary controls.

As shown in *Figure 6.1: Overview of the Six Work Zone Components*, work zones can be divided into six components:

1. Advance Information Zone (optional)
2. Advance Warning Area
3. Transition Area (if a lane or shoulder is closed)
4. Buffer Space
5. Work Activity Area
6. Termination Area

Figure 6.1: Overview of the Six Work Zone Components shows a general concept of the work zone components and how they relate to where devices may be located. For traffic control device selection, refer to the appropriate reference layout found in Sections 7 to 19.

Each component is discussed individually in *Section 6.2: Work Zone Components* for one direction of travel. If the work activity affects both directions of travel, the same principles apply.

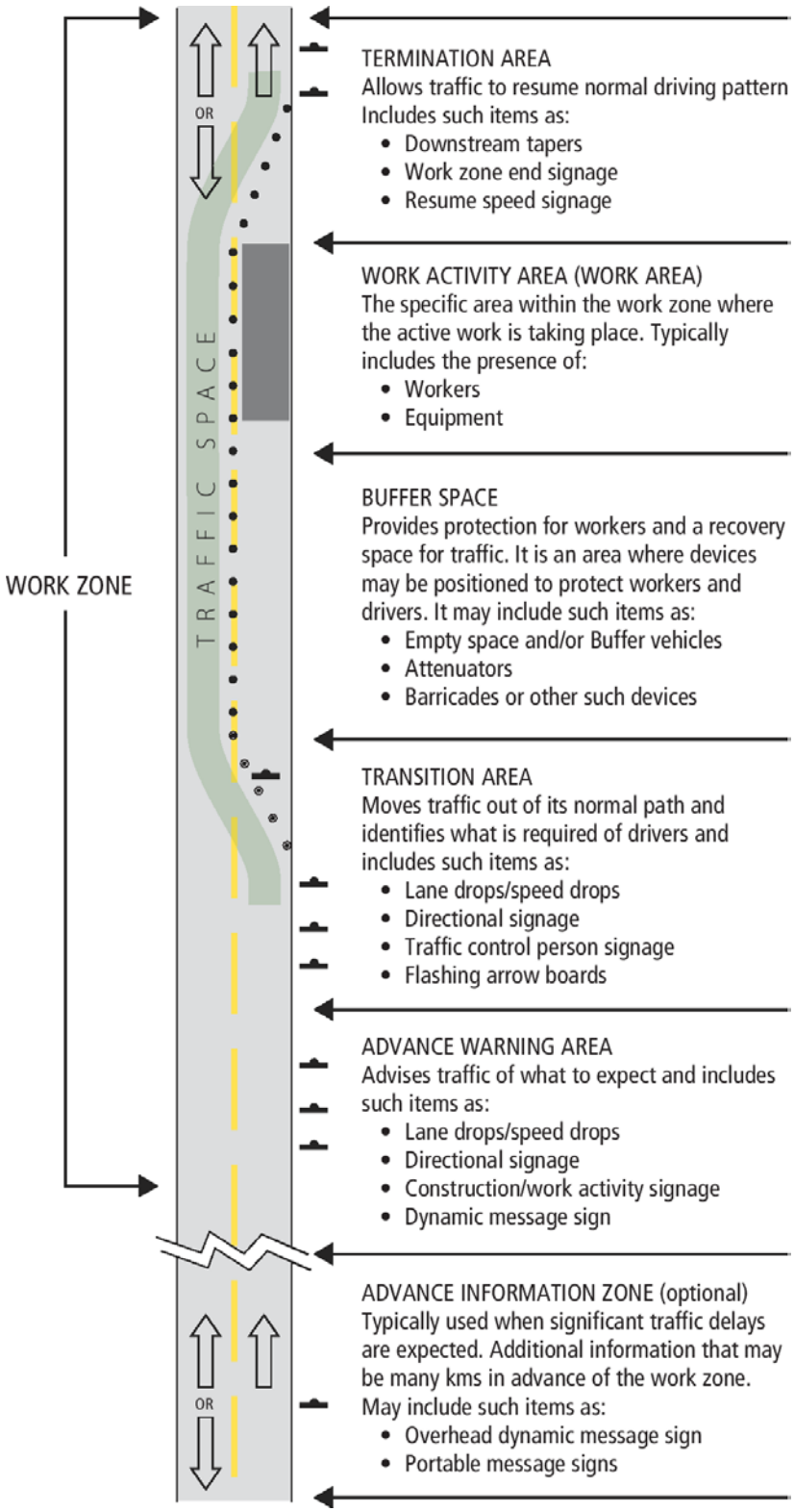


Figure 6.1: Overview of the Six Work Zone Components

6.2 Work Zone Components

6.2.1 Advance Information Zone (optional)

The Ministry has strategically placed overhead Dynamic Messaging Signs (DMS) throughout the province. These can be utilized to provide greater information to travellers, particularly when there are significant impacts or delays. Portable message signs may also be used to provide this advanced information.

The Project Manager would work with the Transportation Management Centre of British Columbia (TMCBC) to post messages on the overhead DMS.

6.2.2 Advance Warning Area

The advance warning area is the area where road users are informed of what to expect ahead. The length of the advance warning area from the first sign to the beginning of the transition area should provide drivers with enough time to adjust their driving patterns safely and appropriately.

The number of traffic control devices in an advance warning area may vary from a single sign or a rotating/flashing light on a vehicle to a series of signs and the use of a portable dynamic message sign (DMS).

The required length of the advance warning area increases with the roadway speed so that road users may more readily perceive and respond to the work condition ahead. Sight distances should be sufficient for drivers to see the situation, interpret what they are seeing, decide what to do, and do what needs to be done (see *Section 2.3.3: Manage Driver Sight Distances*).

Advance warning signs are generally not required when the work activity area and access to it are entirely off the roadway and shoulder, and the work does not interfere with traffic, but they should be used when any traffic flow problems or conflicts could possibly occur.

6.2.3 Transition Area and Tapers

Transition Area: The transition area is the area in which vehicles are channelized/ redirected from their normal path of travel into a new path in order to move around the work activity area. This movement of traffic is achieved by using channelizing devices and directional signs in the tapers that are used to close lanes.

The transition area should be obvious to road users, with the correct path clearly identified with pavement markings and/or channelizing devices so that drivers will not attempt to follow the normally travelled path.

For moving work operations, the transition area moves with the work activity area. A shadow vehicle may be used to warn traffic and/or guide traffic into the proper lane (see *Section 4.11.5: Shadow Vehicles*).

Tapers: Tapers are used in both the transition and termination areas, and are created with a series of channelizing devices or pavement markings placed to move traffic out of—or back into—its normal path. Adjustments to standard taper lengths may be necessitated by the presence of access/egress points and other site constraints. Five different tapers are used within work zones.